

Multiple Choice (10 marks)

1. Consider $x(t)$ to be given as, $x(t) = 10 \cos(20\pi t - \pi/4) - 5 \cos(50\pi t)$ What is the average value of $x(t)$ over one period?
2. 30
3. 20
4. 40
5. 50
6. 35

Which types of functions grow the slowest?

1. $O(N)$
2. $O(\log \log N)$
3. $O(N!)$
4. $O(\log N)$
5. $O(N^N)$

A 3-way, set-associative cache is

1. only effective when more than 3 processes are running alternately on the processor
2. possible only with write-through
3. one in which each main memory word can be stored at any of 4 cache locations
4. possible only with write-back
5. effective only if 3 or fewer processes are running alternately on the processor

Let $x = 8$. What is $x \gg 1$ in Python 3?

1. 5
2. 3
3. 0
4. 8
5. 4

In Python 3, `b = [11,13,15,17,19,21]; print(b[::2])` outputs what?

- 13,17,21
- 13,15,17
- 11,15
- 11,13,15,19
- 11,15,19

True / False (10 marks)

Answer True or False

6. True or False: An excessively large step size can cause the training loss to increase with the number of iterations.
7. True or False: In a simple planar graph with 30 vertices, the maximum number of edges is 84, assuming no self-loops.
8. True or False: A convex polygon with 26 faces and 39 edges has 15 vertices according to Euler's formula.
9. True or False: The TCP connect scan is designed to complete the TCP handshake for port scanning.
10. True or False: The use of weak classifiers in bagging helps prevent overfitting by averaging their predictions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the first repeated character in a given string.
12. Write a function to merge multiple sorted inputs into a single sorted iterator using heap queue algorithm.

End of Paper